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Back-end-of-line thin film resistor

Abstract

Method and resistive structure is provided herein. The resistive structure includes a semiconductor substrate comprising one or more circuit elements and a first interconnect layer disposed on the substrate. The first interconnect layer is between a resistive layer and the semiconductor substrate. A dielectric layer is disposed between the first interconnect layer and the resistive layer. A via extending through the dielectric layer forms an electrical connection between the first interconnect layer and the resistive layer.

Inventors: Li; Baozhen (South Burlington, VT), Yang; Chih-Chao (Glenmont, NY), Dutta; Ashim (Clifton Park, NY), Zhou; Huimei (Albany, NY)

Applicant: International Business Machines Corporation (Armonk, NY)

Family ID: 86768954

Assignee: International Business Machines Corporation (Armonk, NY)

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Primary Examiner: Geyer; Scott B

Attorney, Agent or Firm: Patterson + Sheridan, LLP

Background/Summary

BACKGROUND

(1) The present disclosure relates to back-end-of-line (BEOL) resistors, and more specifically, to BEOL resistive structures having enhanced stability and enhanced thermal dissipation capability.

(2) Thin film resistors are widely used in integrated circuits. As technology scales to smaller interconnects, resistance can increase rapidly. Generally, for new technology nodes, resistors are moved from middle-of-line to BEOL. The resistor can be situated in a structure having a

semiconductor substrate with fins coupled to the substrate, such as in a fin field-effect transistor patterned by extreme ultraviolet lithography.

SUMMARY

(3) According to an embodiment of the present disclosure, a resistive structure is provided herein. The resistive structure includes a semiconductor substrate having one or more circuit elements and a first interconnect layer disposed on the substrate. The first interconnect layer is between a resistive layer and the semiconductor substrate. A dielectric layer is disposed between the first interconnect layer and the resistive layer. A contact via extending through the dielectric layer forms an electrical connection between the first interconnect layer and the resistive layer.

(4) According to an embodiment, a method for forming a resistive structure is provided including depositing a first interconnect layer on a substrate having one or more circuit elements, depositing a cap layer over the first interconnect layer, depositing a resistive layer over the cap layer, and etching one or more openings through the resistive layer and the cap layer. The one or more openings extend to the first interconnect layer. The method further includes filling the one or more openings with a conductive material to form contact vias. The resistive layer contacts the one or more contact vias to form an electrical connection with the first interconnect layer.

(5) According to some embodiments, a method for forming a resistive structure is provided including depositing a first interconnect layer on a substrate having one or more circuit elements. The method includes depositing a cap layer over the first interconnect layer, etching one or more openings through the cap layer and filling the one or more openings to form one or more studs. The one or more studs extend through the cap layer to the first interconnect layer. The method further includes depositing a resistive layer over the cap layer. The resistive layer contacts the one or more studs to form an electrical connection with the first interconnect layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts a flow diagram for a method of forming a thin film resistive structure according to some embodiments.

(2) FIG. 2 depicts a structure with a resistive layer disposed over a capping layer and interconnect layer according to some embodiments.

(3) FIG. 3 depicts a structure with openings formed therein according to some embodiments.

(4) FIG. 4 depicts a structure with contact vias according to some embodiments.

(5) FIG. 5 depicts a thin film resistive structure according to some embodiments.

(6) FIG. 6 depicts a flow diagram for a method of forming a thin film resistive structure according to some embodiments.

(7) FIG. 7 depicts a structure having a capping layer disposed over a patterned interconnect layer according to some embodiments.

(8) FIG. 8 depicts a structure having studs disposed through a capping layer disposed over a patterned interconnect layer according to some embodiments.

(9) FIG. 9 depicts a structure having a resistive layer disposed over studs according to some embodiments.

(10) FIG. 10 depicts a thin film resistive structure according to some embodiments.

(11) FIG. 11 depicts a thin film resistive structure having studs through a resistive layer according to some embodiments.

DETAILED DESCRIPTION

(12) The present disclosure is directed to back-end-of-line (BEOL) thin film resistor (TFR) structures disposed on a substrate. The resistor forms a portion of an integrated circuit. Compared to middle-of-line structures, BEOL TFRs are spaced a greater distance away from the substrate.

The resistor is surrounded by material having lower thermal conductivity, such as a dielectric constant (K) of about 3.5 or less, such as about 2.7 or less, such as about 2.5 or less. Relative to middle-of-line resistors, reduced thermal dissipation is observed in conventional BEOL resistors. Conventional solutions to the reduced thermal dissipation include limiting the electrical current passing through the resistor, setting an exclusion zone for metal wiring, or derating the electromigration limit for the vias and lines connected to (or proximate to) the TFR. Each of these solutions limit or compromise the performance of the final device. The resistors described herein provide the benefits of BEOL transistors without affecting the performance of the devices, such as limiting current for electromigration, while enhancing thermal dissipation. In particular, contact vias are formed in the resistive structures extending at least partially through a resistive layer of the structure, or at least partially below the resistive layer of the structure. Conventional contact vias are typically disposed above resistive layers. A short length effect for the connecting conductor wires is observed for electromigration within resistors of the present disclosure. The resulting electromigration limit is less sensitive to temperature rise. Moreover, the metal connection line disposed below the resistive layer is at a lower level and enhances thermal dissipation to the substrate.

(13) Additionally, when high amounts of current passes through conventional resistors, high amounts of joule heating can occur, especially at a bottom of the contact vias, which can accelerate electromigration and form a void. Formation and growth of voids can affect device performance by opening the circuit of the device. As used herein, the term “electromigration” refers to migration of metal atoms gradually moving through a conductor due to momentum transfer between conducting electrons and diffusing metal atoms, which can cause electrical openings or shorting.

Electromigration can be reduced at the expense of current allowance. It has been discovered that resistors of the present disclosure are substantially free of voids and more resistant to high temperatures.

(14) FIG. 1 depicts a flow diagram for a method **100** of forming a thin film resistive structure according to some embodiments and is illustrated. The method **100** includes, at block **102**, depositing a first interconnect layer over a substrate, at block **104**, depositing a cap layer over the first interconnect layer, at block **106**, depositing and patterning a resistive layer over the cap layer, and at block **108**, etching one or more openings through the resistive layer and the cap layer. The one or more openings extend through the resistive layer and the cap layer to the first interconnect layer. At block **110**, the one or more openings are filled with a conductive material. The filled openings can be referred to as contact vias, metal connections, or studs. FIGS. 2-5 illustrate one example of performing the method **100**.

(15) FIG. 2 depicts a structure **200** corresponding to block **102** of the method **100**. Specifically, FIG. 2 depicts the structure **200** after depositing a first interconnect layer **204** over a substrate **202**. The first interconnect layer **204** can be deposited using any suitable method and can include conductive materials, such as copper, cobalt, ruthenium, aluminum, combinations thereof, or alloys thereof. Other conductive materials, such as other metals are also contemplated. In one embodiment, the first interconnect layer **204** is a first interconnect layer disposed closest to the substrate **202**. Alternatively, in one embodiment, the first interconnect layer is disposed above one or more additional interconnect layers (not shown).

(16) In one embodiment, the substrate **202** is a semiconductor substrate that contains transistors, and other circuit elements of an integrated circuit. Thus, the process illustrated in FIGS. 2-5 is a BEOL process that occurs after the circuitry in the substrate **202** (e.g., transistors) has been formed.

(17) The first interconnect layer **204** is patterned and etched using any suitable method. A cap layer **206** is formed over the first interconnect layer **204**, which corresponds to block **104** of the method **100**. In some embodiments, the cap layer **206** is composed of a dielectric material. In some embodiments, the cap layer **206** is a protective cap such as silicon nitride, silicon carbide (e.g., nitrogen doped silicon carbide), aluminum nitride, oxy-nitride, or combination(s) thereof. In some

embodiments, the cap layer **206** is a silicon-containing material. A resistive layer **208** is deposited and patterned over the cap layer **206** which corresponds to block **106** of the method **100**. The resistive layer **208** is composed of SiCr, TiN, TaN, or other resistive materials.

(18) FIG. **3** depicts a structure **300** corresponding to block **108** of the method **100**. Specifically, FIG. **3** depicts the structure **300** after forming openings **302** by etching through an interlayer dielectric (ILD) material **210**, the resistive layer **208**, and the cap layer **206**. Additional openings **304** can also be formed through the ILD material **210** and the cap layer **206** at portions of the structure that do not include the resistive layer **208**. Without being bound by theory, it is believed that because of the hardness of the resistive layer **208**, the width of the openings is reduced at the upper surface of the resistive layer **208**, such that the width of the opening in the ILD material **210** (e.g., width **308**) is wider than the width of the opening in the resistive layer **208** (e.g., width **306**). The reduced width is due to the faster etch rate of the ILD material **210** relative to the resistive layer **208**. It has been discovered that the increased width **308** within the ILD material **210** enables forming a structure having lower contact resistance relative to a structure having a uniform width (e.g., width **306**) within the ILD material **210** and the resistive layer **208**. Although the width **306** of the opening in the resistive layer **208** and the cap layer **206** is uniform in width in FIG. **3**, it is also contemplated that the width may vary. For example, in some embodiments, the width **306** of the opening in both the resistive layer **208** and the cap layer **206** decreases from an upper surface of the resistive layer **208** to a lower surface of the cap layer **206**.

(19) In some embodiments, the ILD material **210** is a dielectric material having a low dielectric constant, such as about 3.5 or lower. In some embodiments, the ILD material **210** is a low K dielectric material, such as SiC_xO_yH_z. In some embodiments, the ILD material **210** is deposited using chemical vapor deposition (CVD), spin on, or a combination thereof.

(20) FIG. **4** depicts a structure **400** corresponding to block **110** of the method **100**. Specifically, FIG. **4** depicts the structure **400** after filling the openings (e.g., **302** of FIG. **3**) with a conductive metal to form contact vias **403**. Each contact via **403** includes a first portion **402** and a second portion **404**. Each contact via **403** is electrically connected an interconnect portion (e.g., **406A**, **406B**). In an embodiment, the contact vias **403** include a single first portion **402** and a single second portion **404** electrically connected to a single interconnect portion **406B**, as illustrated. In an alternative embodiment, the contact vias **403** include multiple first portions **402** and multiple second portions **404** electrically connected to a single interconnect portion **406A**, as illustrated. Moreover, in an embodiment, the structure **400** includes at least one set of contact vias **403** having multiple first and second portions **402**, **404** connected to a single interconnect portion **406A** and at least one contact via **403** having each first and second portion **402**, **404** electrically connected to a single interconnect portion **406B**, as illustrated. In some embodiments, the contact via **403** extends through an entire thickness of the resistive layer **208**.

(21) Additional contact vias and interconnect portions, such as contact via **405** and interconnect portion **410** can be formed by filling openings (e.g., opening **304** shown in FIG. **3**) at the portions of the structure that do not include the resistive layer **208**. The additional contact vias **405** and interconnection portions **410** form portions of additional circuits and connects on the substrate. The interconnect portions **406A**, **406B**, **410** together form a second interconnect layer **406**. In some embodiments, the contact vias **403** electrically connect the first interconnect layer **204** with the second interconnect layer **406** disposed on an opposing side of the resistive layer **208**. In some embodiments, the conductive material of the contact vias are composed of the same material as the first interconnect layer **204**. Alternatively, the conductive material of the contact vias are a different material from the first interconnect layer **204**. In some embodiments, the conductive material of contact vias is copper, cobalt, tungsten, alloys thereof, or combinations thereof. Other conductive materials are also contemplated such as other metals and alloys. Each interconnect portion **406A**, **406B** of the second interconnect can be connected to a portion of the first interconnect layer through the second portions **404** and first portions **402** of the contact vias **403**. At least one of the

interconnect portions **406A**, **406B** can have a length **407** less than a length **409** of an interconnect portion **204A**, **204B** of the first interconnect layer **204** that is electrically connected to the at least one interconnect portion **406A**, **406B** of the second interconnect layer **406**. For example, interconnect portion **406A** can have a length **407** that is less than a length **409** of the interconnection portion **204A** of the first interconnect layer **204**. In some embodiments, a width of each of the interconnect portions **406A**, **406B** in a Z-direction is about 0.01 μm or wider, such as 0.1 μm to about 1 μm . In some embodiments, the length **407** of each of the interconnect portions **406A**, **406B** is about 0.03 μm to about 1 μm .

(22) In some embodiments, the interconnect portions **406A**, **406B** are not necessary for electrical conduction; however, the interconnection portions **406A**, **406B** are formed to cover the contact vias **403**. By way of example, for Cu dual damascene process to form a first portion **402** of the contact via **403**, a small metal island or button (e.g., second interconnect portion **406B**) is formed to cover the contact via **403**. Without being bound by theory, it is believed that due to the small volume of metal (such as Cu) in an interconnect portion **406A**, **406B**, the electromigration reliability of the contact vias **403** connecting the TFR is significantly enhanced. In some embodiments, the contact vias **403** is immortal to electromigration.

(23) FIG. 5 depicts an example current flow path for the structure **400** depicted in FIG. 4. Without being bound by theory, it is believed for structures such as the structure **400** having all electrical connections through the first interconnect layer **204** and having small interconnect portions **406A**, **406B** of the second interconnect layer **406**, there is substantially no driving force for current to flow to the second interconnect layer **406**. It is further believed that when electrons flow upwards to the resistive layer **208** through contact via **403**, such as along current path C1, electromigration resistance for contact via **403** is significantly enhanced due to the lack of current flow through the small interconnect portions **406A**, **406B**. In some embodiments, such as in the left end of the structure **400**, electrons flow downwards from contact via **403** to the first interconnect layer **204** below, such as along current path C2. The contact via **403** having current path flowing downward is at an accumulation mode in which there is no or substantially no electromigration for the contact via **403**. In some embodiments, to enhance the first interconnect layer **204** electromigration resistance, individual contact vias **403** individually connected to individual interconnect portions (e.g., **204B**) of the first interconnect layer **204** can be implemented. Each interconnect portion **204A** can have a short length to enhance electromigration resistance of the first interconnect layer **204**.

(24) FIG. 6 depicts a flow diagram for a method **600** of forming a thin film resistive structure according to some embodiments. The method **600** includes, in block **602**, depositing a first interconnect layer over a substrate. In block **604**, a cap layer is formed over the first interconnect layer, and in block **606**, one or more openings are etched through the cap layer. The one or more openings extend through the cap layer to the first interconnect layer. The method further includes, in block **608**, filling the one or more openings to form studs. The method further includes, at block **610**, depositing a resistive layer over the cap layer. FIGS. 7 to 11 illustrate one example of performing the method **100**.

(25) FIG. 7 depicts a structure **700** corresponding to blocks **602** and **604** of method **600**. Specifically, FIG. 7 depicts the structure **700** having a substrate **702**, such as a semiconductor substrate. In one embodiment, the substrate **702** is similar to the substrate **202** described relative to FIGS. 2 to 5. Each of a first interconnect layer **704** and a cap layer **706** is formed using similar processes and compositions used for the first interconnect layer **204** and cap layer **206** described relative to FIGS. 2 to 5.

(26) FIG. 8 depicts a structure **800** corresponding to blocks **606** and **608** of method **600** of FIG. 6. Specifically, FIG. 8 depicts the structure **800** after etching one or more openings into the cap layer **706** to the first interconnect layer **704**, as described in block **606** of FIG. 6. The openings are filled with a conductive material to form studs **708**, which corresponds to block **608** of FIG. 6. In some embodiments, the studs **708** are composed of tungsten, copper, cobalt, ruthenium, or a combination

thereof, or their alloys. An upper portion of the studs **708** along an upper surface of the cap layer **706** can be planarized, such as by chemical mechanical polishing (CMP). The top surface of the studs **708** are substantially coplanar with the top surface of the cap layer **706**. FIG. **9** depicts a structure **900** corresponding to block **610** of method **600** of FIG. **6**. Specifically, FIG. **9** depicts the structure **900** after forming a resistive layer **710** over the studs **708** and cap layer **706**. In one embodiment, the resistive layer **710** is formed using similar processes and compositions used for the resistive layer **208** described relative to FIGS. **2** to **5**.

(27) FIG. **10** depicts a structure **1000** after depositing an interlayer dielectric (ILD) material **1006** over the structure **900**. The ILD material **1006** can be any material described relative to the ILD material **210** depicted in FIGS. **2** to **5**. One or more openings can be formed at portions of the structure **1000** that do not include the resistive layer **710**. The openings can be filled to form a contact via **1002** and an interconnect portion **1004**. The interconnect portion **1004** can be electrically connected to the first interconnect layer **704** through the contact via **1002**. The interconnect portion **1004** and contact via **1002** form portions of other circuits disposed on the substrate **702** that are not part of the resistive structures.

(28) Similar to the current paths described in FIG. **5**, the electrons flow along current path C3 from the first interconnect layer **704**, through studs **708** to upper surfaces of the studs **708**, and through the resistive layer **710**. The resistive layer **710** interfaces the studs **708** at an interface that is coplanar with the upper surfaces of the studs **708**. In some embodiments, the electrons flow along current path C4 from the resistive layer **710**, downward through the studs **708** through the interface between the studs **708** and resistive layers **710**, and through the first interconnect layer **1104**. The current paths C3, C4 depicted in FIG. **10** is different from conventional current paths which typically flow from above resistive layer of conventional structures.

(29) Other embodiments are also contemplated such as the structure **1100** shown in FIG. **11** having a substrate **1102**, first interconnect layer **1104**, cap layer **1106**, and studs **1110** that extend through the resistive layer **1108** from a first surface to a second surface of the resistive layer **1108**. The process for forming the studs **1110** is similar to method **600** of FIG. **6**, except after block **604**, a resistive layer **1108** is formed over the cap layer **1106**. The one or more openings are etched through the resistive layer **1108** and the cap layer **1106**. The openings are filled to form the studs **1110**. The upper surface of the studs **1110** can be planarized using chemical mechanical polishing (CMP) methods such that the upper surface of the studs **1110** are substantially coplanar with the upper surface of the resistive layer **1108**.

(30) In some embodiments, the structure **1100** includes interconnect portions **1112** disposed in portions of the structure **1100** that are not part of the resistive structure. The interconnect portions **1112** are formed similar to the interconnect portions **1004** described relative to structure **1000** of FIG. **10**. Similar to FIG. **10**, the electrons flow along current path C5 through the first interconnect layer **1104**, through sidewalls of the studs **1110**, and through the resistive layer **1108**. In some embodiments, the electrons flow along current path C6 through the resistive layer **1108**, through a sidewall of the stud **1110**, downward through the studs **1110**, and through the first interconnect layer **1104**. The current paths C5, C6 depicted in FIG. **11** are different from conventional current paths which typically flow from above resistive layer of conventional structures. Without being bound by theory, it is believed that the structure **1100** shown in FIG. **11** has lower contact resistance than the structure **1000** shown in FIG. **10**.

(31) In some embodiments, the portions of the studs or contact vias that extend through the cap layer or resistive layer are substantially uniform in length. In some embodiments, the width of the studs or contact vias that extend through the cap layer or resistive layer taper to a more narrow width at portions of the contact via or studs closer to the substrate. For example, the contact via has a width that is tapered from an uppermost portion of the contact via to a lowermost portion of the contact via contacting the first interconnect layer.

(32) In conventional resistive structures, contact vias are formed above the resistive layer and

contact only a top surface of the resistive layer. In contrast, the resistors provided herein include more contact interfaces between the via and the resistive layer. For example, the contact vias include sidewall contact with the resistive layer and top contact. It has been discovered that the resistors described herein demonstrated enhanced resistor performance and higher current allowance through the resistor. Additionally, the resistor demonstrated better electromigration reliability, better joule heating tolerance, and better contact resistance. In some embodiments, each of the contact vias described herein include a liner disposed at an interface of the contact vias and the first interconnect. The liner reduces metal atom migration, such as copper migration there through and prevents void formation within the contact vias. As used herein, the “liner” can be composed Ta, TaN, or a combination thereof.

(33) The resistors described herein include contact vias formed in the resistive structures that extend at least partially through a resistive layer of the structure, or at least partially below the resistive layer of the structure. A short length effect is observed for electromigration for the contact vias connecting to the resistors of the present disclosure. The resulting electromigration limit is not sensitive to temperature rise. Moreover, the metal connection line disposed below the resistive layer is at a lower level and enhances thermal dissipation.

(34) In the preceding, reference is made to embodiments presented in this disclosure. However, the scope of the present disclosure is not limited to specific described embodiments. Instead, any combination of the features and elements, whether related to different embodiments or not, is contemplated to implement and practice contemplated embodiments. Furthermore, although embodiments disclosed herein may achieve advantages over other possible solutions or over the prior art, whether or not a particular advantage is achieved by a given embodiment is not limiting of the scope of the present disclosure. Thus, the aspects, features, embodiments and advantages discussed herein are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s). Likewise, reference to “the invention” shall not be construed as a generalization of any inventive subject matter disclosed herein and shall not be considered to be an element or limitation of the appended claims except where explicitly recited in a claim(s).

(35) While the foregoing is directed to embodiments of the present invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A resistive structure comprising: a semiconductor substrate comprising one or more circuit elements; a first interconnect layer disposed on the substrate; a resistive layer, wherein the first interconnect layer is between the resistive layer and the semiconductor substrate; a dielectric layer disposed between the first interconnect layer and the resistive layer; and a contact via extending through the dielectric layer to form an electrical connection between the first interconnect layer and the resistive layer, wherein the contact via extends through at least a portion of the resistive layer.
2. The resistive structure of claim 1, wherein the contact via extends through an entire thickness of the resistive layer and a portion of the contact via disposed above an upper surface of the resistive layer is wider than a portion of the contact via disposed in the resistive layer.
3. The resistive structure of claim 1, wherein a portion of the contact via disposed in the resistive layer and the dielectric layer is substantially uniform in width.
4. The resistive structure of claim 1, wherein a first surface of the contact via contacts a second interconnect layer and a second surface of the contact via contacts the first interconnect layer.
5. The resistive structure of claim 4, further comprising a plurality of contact vias, wherein at least two contact vias contact the second interconnect layer.
6. The resistive structure of claim 1, wherein the contact via is a stud that interfaces the resistive

layer at a resistive layer surface that interfaces the dielectric layer.

7. The resistive structure of claim 6, wherein the stud is a metal stud comprising one or more of tungsten, copper, cobalt, ruthenium, alloys thereof, or combinations thereof.

8. The resistive structure of claim 1, wherein the contact via has a width that is tapered from an uppermost portion of the contact via to a lowermost portion of the contact via contacting the first interconnect layer.

9. The resistive structure of claim 1, wherein the dielectric layer is selected from silicon nitride, nitrogen doped silicon carbide, aluminum nitride, oxy-nitride, or combinations thereof.

10. A method for forming a resistive structure, comprising: depositing a first interconnect layer on a substrate comprising one or more circuit elements; depositing a cap layer over the first interconnect layer; depositing a resistive layer over the cap layer; etching one or more openings through the resistive layer and the cap layer, the one or more openings extending to the first interconnect layer; and filling the one or more openings with a conductive material to form one or more contact vias, wherein the resistive layer contacts the one or more contact vias to form an electrical connection with the first interconnect layer.

11. The method of claim 10, further comprising patterning the first interconnect layer after depositing the first interconnect layer.

12. The method of claim 10, further depositing an interlayer dielectric (ILD) material over the resistive layer prior to etching the one or more openings.

13. The method of claim 12, wherein filling the one or more openings further comprises forming first portions of the one or more contact vias, and second portions of the one or more contact vias disposed over the first portions of the one or more contact vias and in the ILD material.

14. The method of claim 13, wherein the second portions comprise a width greater than a width of the first portion of the one or more contact vias disposed in the resistive layer.

15. The method of claim 10, wherein the one or more contact vias electrically connect the first interconnect layer to a second interconnect layer disposed on an opposing side of the resistive layer.

16. The method of claim 10, wherein the first interconnect layer is composed of copper, cobalt, ruthenium, tungsten, aluminum, alloys thereof, or combinations thereof.
